Assignment 13.4

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Batch: 44

Task 1: Refactoring Data Transformation Logic

Code:

```
raw_values = input("Enter numbers separated by spaces: ")
values = [int(v) for v in raw_values.split()]
doubled = [v * 2 for v in values]

print(doubled)
```

Input:

```
Enter numbers separated by spaces: 2 4 6 8 10
```

Output:

```
[4, 8, 12, 16, 20]
```

Task 2: Improving Text Processing Code Readability

Code:

```
words = ["Refactoring", "with", "AI", "improves", "quality"]
message = " ".join(words)
print(message)
```

Output:

```
Refactoring with AI improves quality
```

Task 3: Safer access to Configuration Data

Code:

```
config = {"host": "localhost", "port": 8080}

timeout = config.get("timeout")
if timeout is not None:
    print(timeout)
else:
    print("Default timeout used")
```

Output:

```
Default timeout used
```

Task 4: Refactoring Conditional Logic for Scalability

Code:

```
action = "divide"
x, y = 10, 2

operations = {
    "add": lambda a, b: a + b,
    "subtract": lambda a, b: a - b,
    "multiply": lambda a, b: a * b,
    "divide": lambda a, b: a / b,
}

operation = operations.get(action)
result = operation(x, y) if operation else None

print(result)
```

Output:

```
5.0
```

Task 5: Simplifying Search Logic in Collections

Code:

```
inventory = ["pen", "notebook", "eraser", "marker"]
found = "eraser" in inventory

print("Item Available" if found else "Item Not Available")
```

Output:

```
Item Available
```